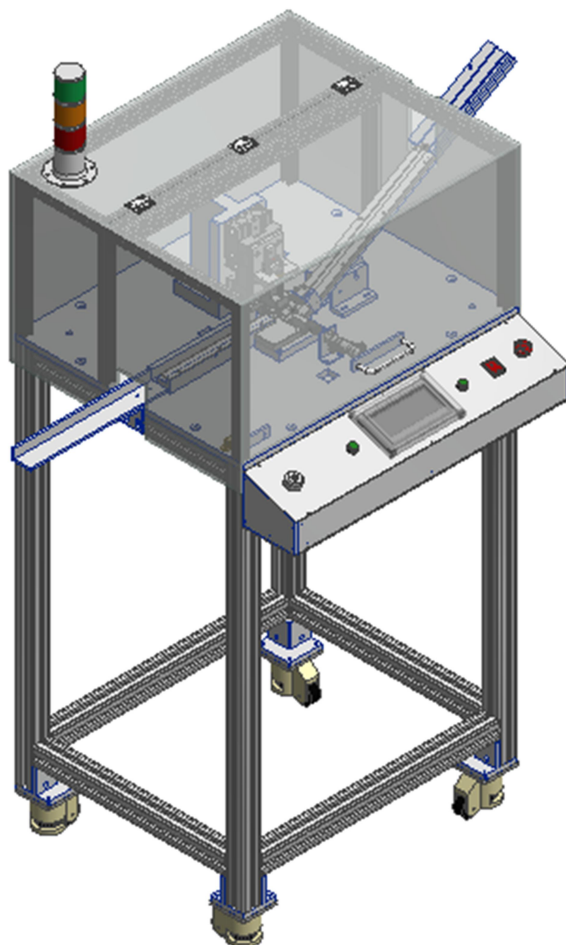




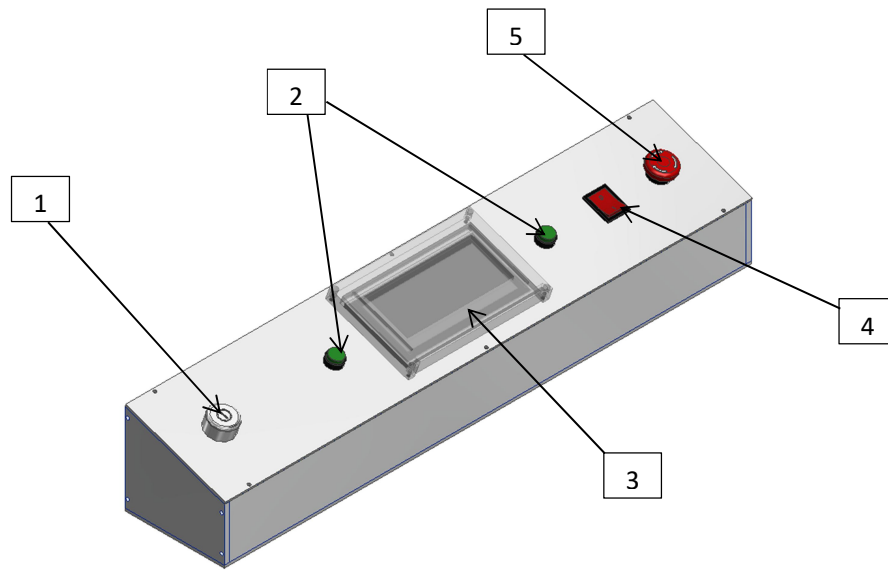
PANASONIC AUTO DIODE
FORMING MACHINE



OPERATION MANUAL

OPERATION MANUAL

MACHINE OPERATION PANEL



OPERATION PANEL

No	Function	Description
1	MAINTENANCE KEY SWITCH	TURN TO ENABLE OPERATION OF MACHINE
2	START BUTTON	PRESS TO START THE MACHINE
3	TOUCH PANEL DISPLAY	PRESS TO SELECT MODE
4	POWER ON SWITCH	TURN ON TO GIVE POWER TO MACHINE
5	EMERGENCY BUTTON	PRESS TO EMERGENCY STOP

MACHINE FRONT PANEL

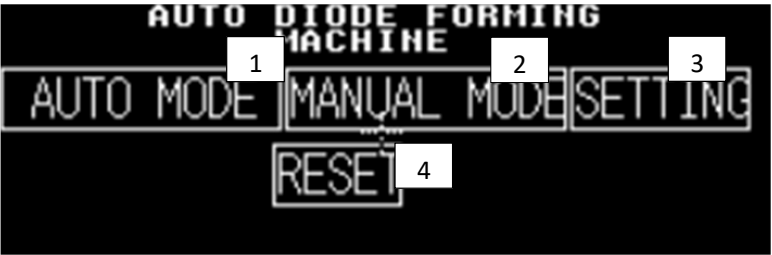


Figure 1 : Main panel

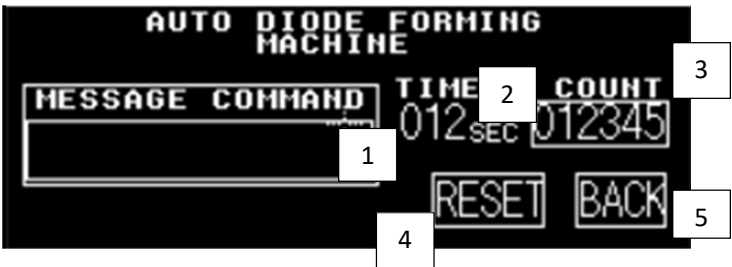
MAIN PANEL

No	Function	Description
1	AUTO MODE	WHEN TURNED ON, THE MACHINE IS IN AUTO MODE AND READY FOR PRODUCTION. WHEN TURNED OFF, THE MACHINE IS IN DEBUGGING OR REPAIRING MODE.
2	MANUAL	TO ENTER IO SETTING MODE
3	SETTING	TO ENTER SETTING MODE
4	RESET	Reset all I/O and address value on the PLC and reset the to initial position

MACHINE OPERATION PROCEDURE

1. POWER ON THE MACHINE BY SWITCH ON POWER BUTTON.
2. TURN ON AIR SUPPLY TO THE MACHINE AND MAKE SURE EMERGENCY BUTTON IS IN OFF POSITION.
3. PRESS RESET BUTTON AND MACHINE WILL PERFORM HOMING/RETURN TO INITIAL POSITION ROUTINE.
4. SET MACHINE TO AUTO MODE.
5. LOAD PRODUCT IN MACHINE AND PRESS THE START BUTTON.
6. WHEN PROCESS IS COMPLETED, MACHINE STATUS WILL DISPLAY "MACHINE STANDBY" ONCE AGAIN AND OPERATOR CAN UNLOAD THE PRODUCT FROM THE MACHINE.

AUTO FUNCTION



AUTO SCREEN

No	Buttons	Description
1	MESSAGE COMMAND	Machine status.
2	TIMER	Cycle time Stopwatch
3	RESET	Reset machine to initial position.
4	COUNT	Count Product.
5	BACK	Go to previous page

SETTING FUNCTION



Figure 2 : Setting Screen

SETTING SCREEN

No	Buttons	Description
1	TIMER	Delay while push lock wait x10ms/ second selected
2	RESET COUNTER	Reset the product count.
3	STOP COUNTER	Amount of product counter before stop

MANUAL FUNCTION



Figure 3 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	To check the sensor/input signal.

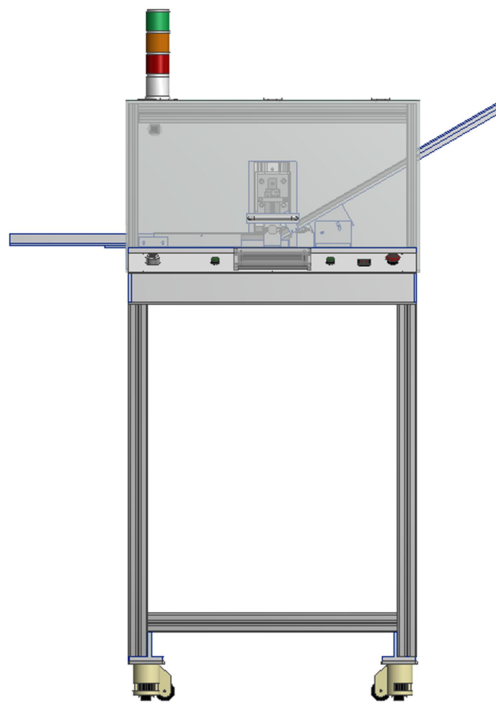
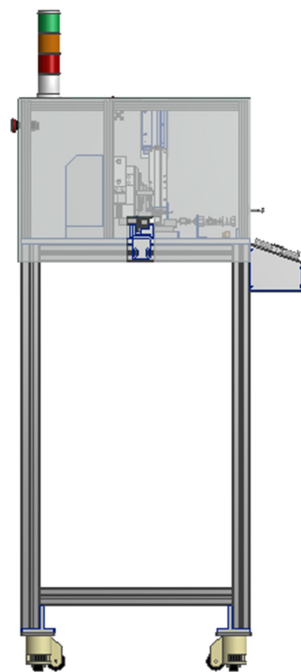


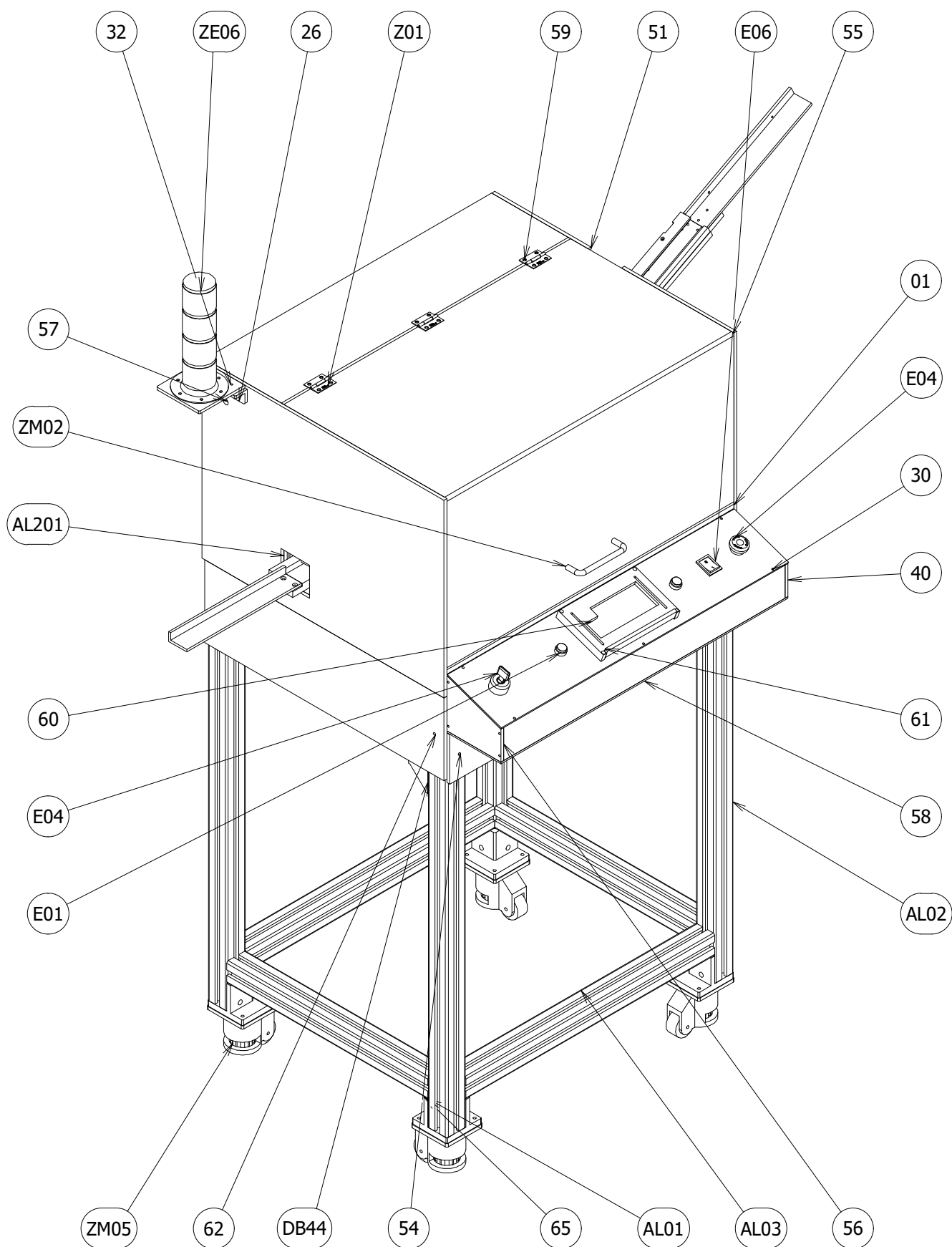
Figure 4 : Output Screen

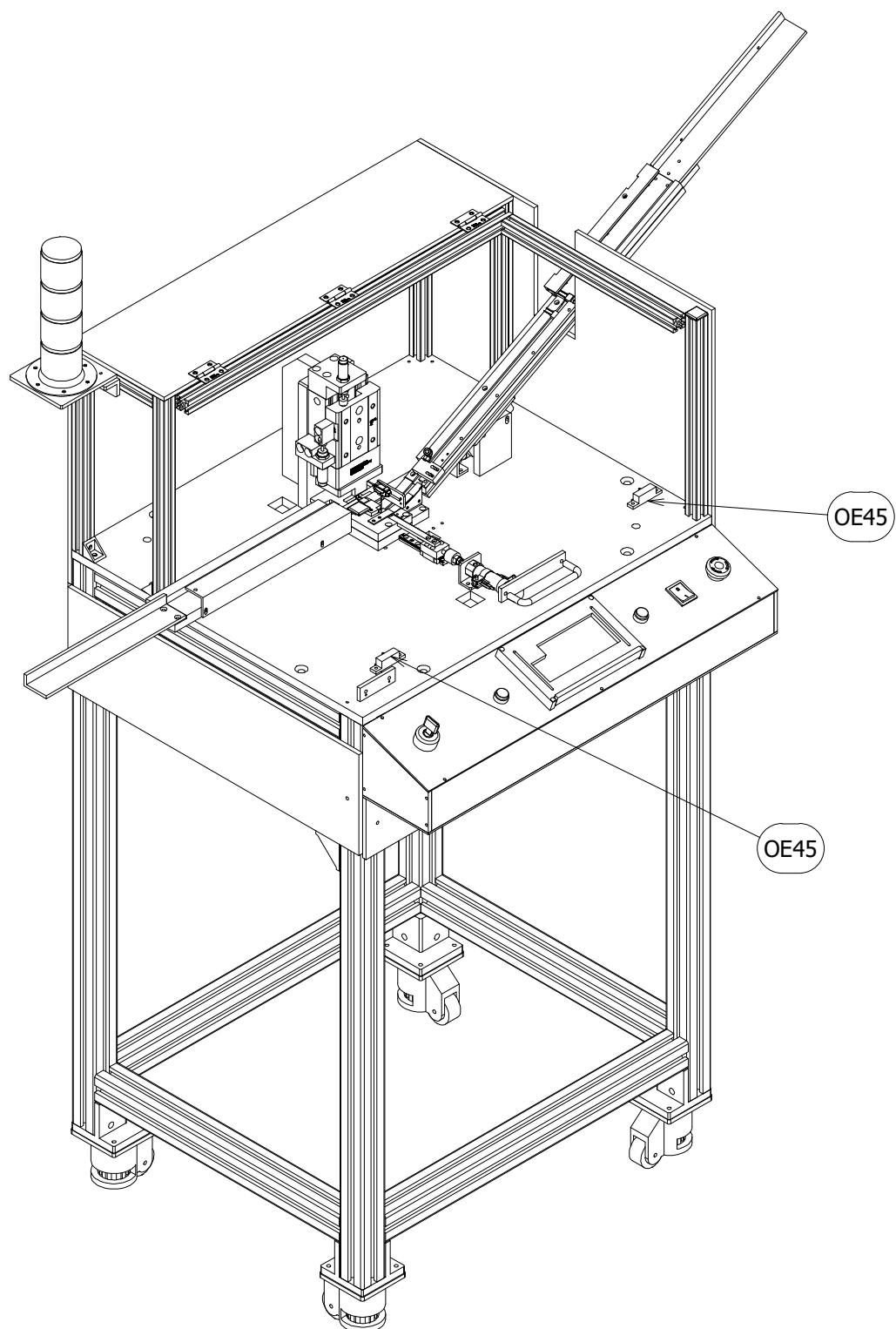
OUTPUT SCREEN

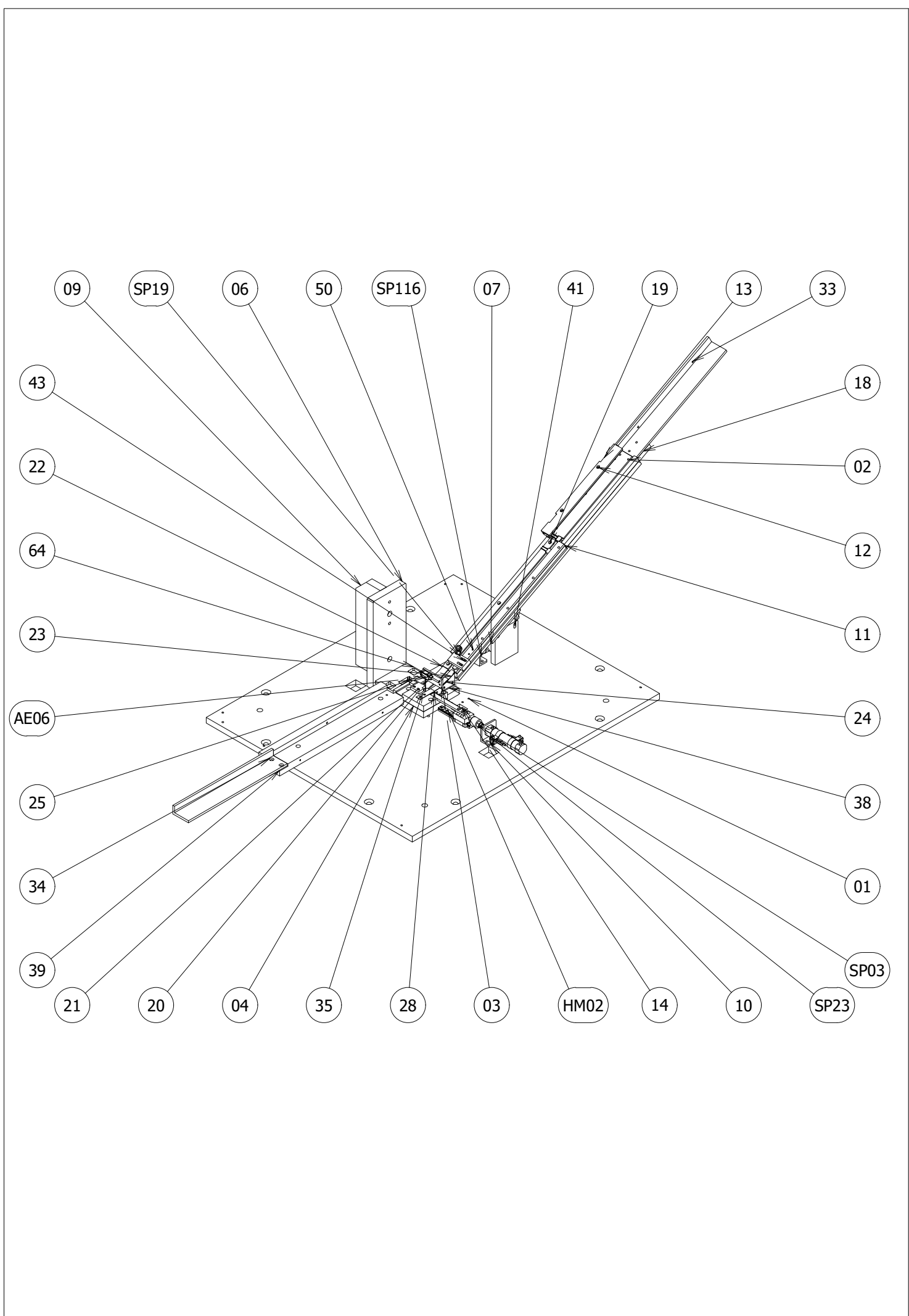
No	Function	Description
1	OUTPUT	Manual trigger output.

MECHANICAL & ELECTRICAL
PART LIST







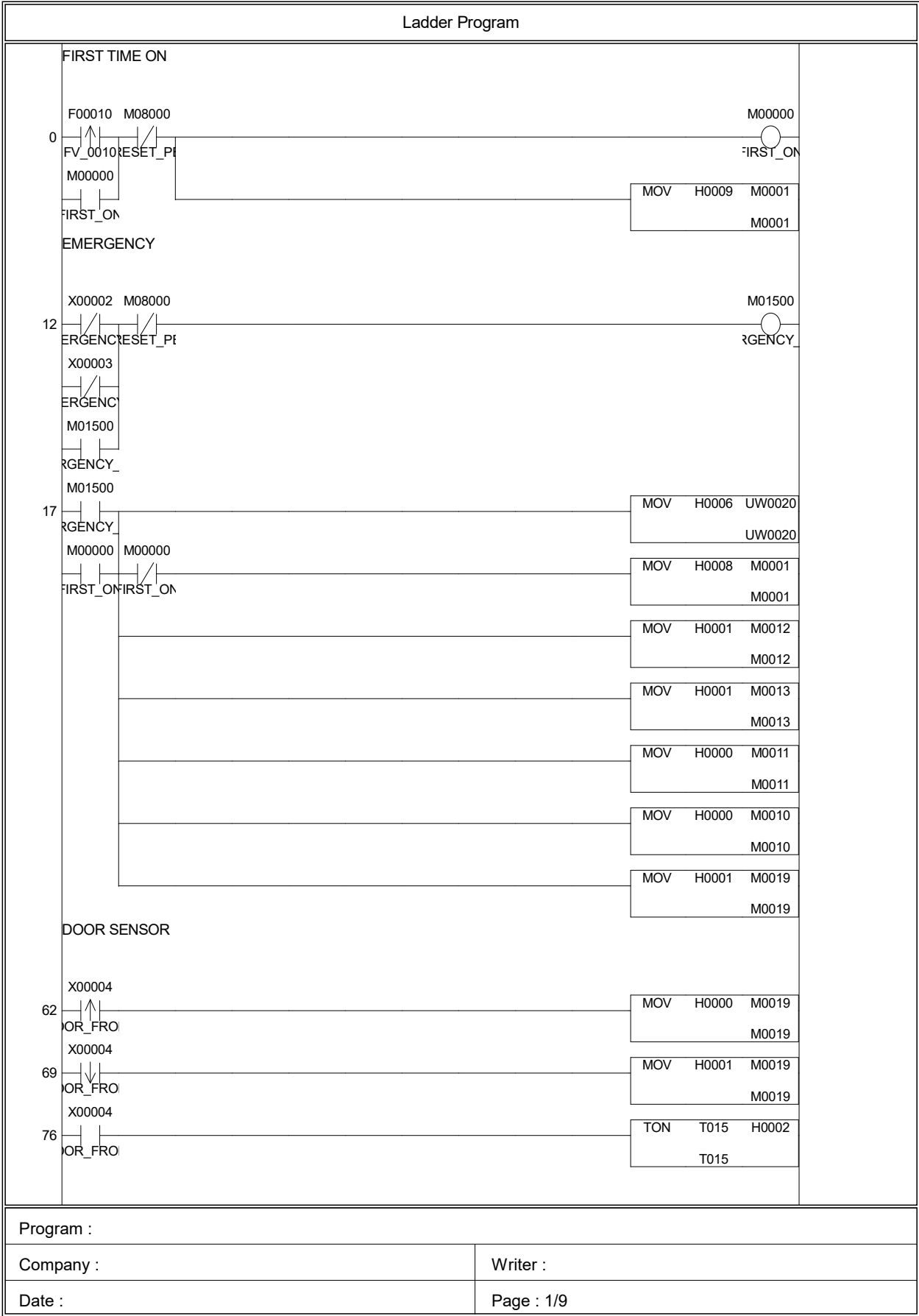


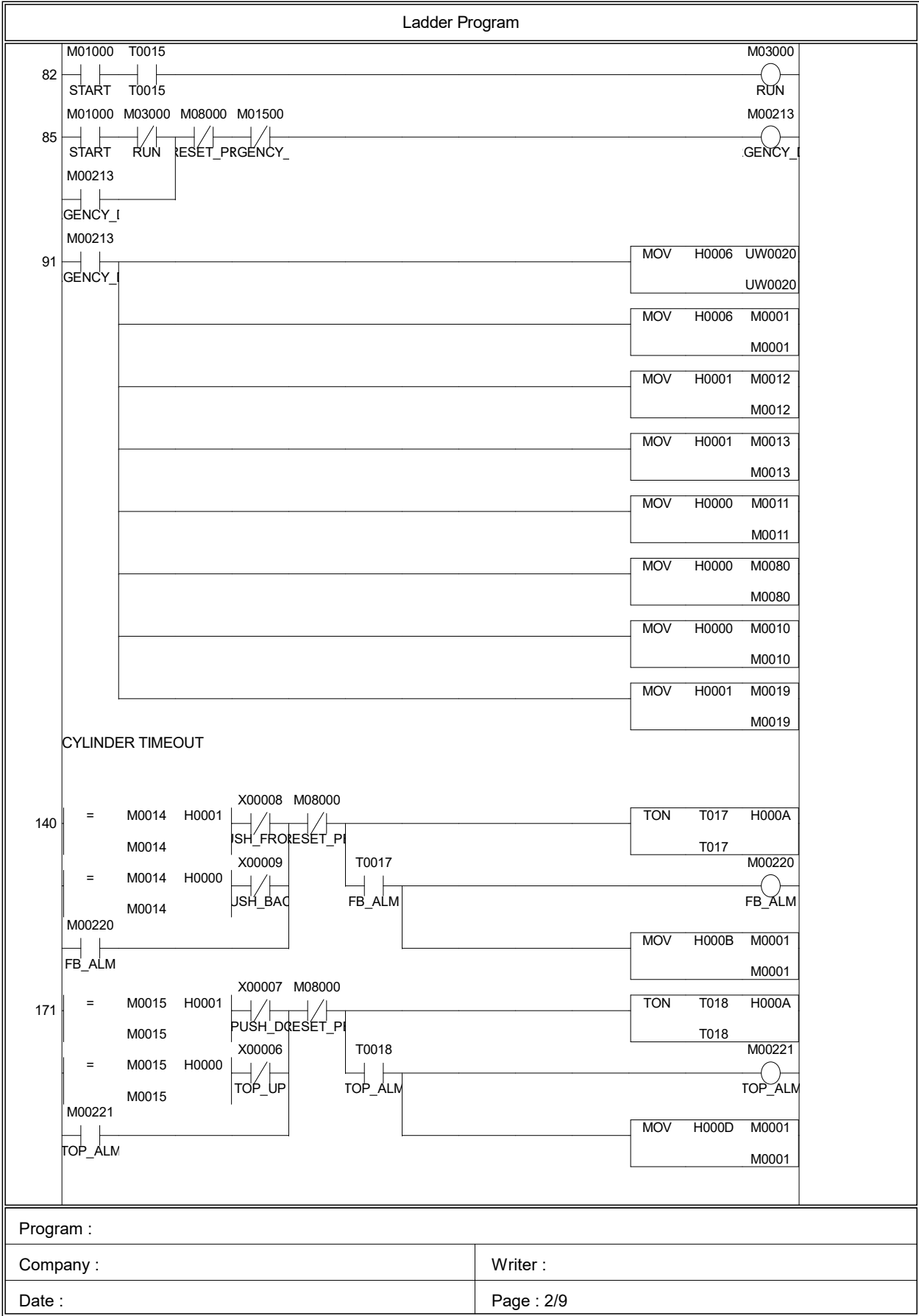
PARTS LIST		
ITEM No	QTY	PART NAME
01	1	BASE TYPE 1
02	1	T GUIDE 4 TYPE 1
03	1	S1
04	1	TX HOLDER 2 TYPE 1
05	1	OP L 2 TYPE 1
06	1	CYL STAND 1 TYPE 1
07	1	R PLATE 2
08	1	CYL STAND 3 TYPE 1
09	1	CYL STAND 2
10	2	CYL PLATE 1 TYPE 2
11	1	TUBE LOCK IP 1
12	1	TUBE LOCK IP 3C
13	1	COVER 2B
14	2	PUSHER FRONT 2
15	2	A BLOCK
16	1	CYL STOPPER 1
17	1	CYL STOPPER 2
18	1	SUPPORT L5
19	1	INPUT BLOCK 1 TYPE 1
20	1	TX INPUT BLOCK 2 TYPE 1
21	1	TX HOLDER 1 TYPE 1
22	1	PUSHER SIDE 2 TYPE 1
23	1	B COVER 1
24	1	B COVER 2
25	1	LEAD PUSHER 1B
26	1	T PLATE 2
27	1	LEAD PUSHER 2B
28	1	PUSHER FRONT 1
30	1	SBOX COVER
31	1	SIDE SB PLATE 1

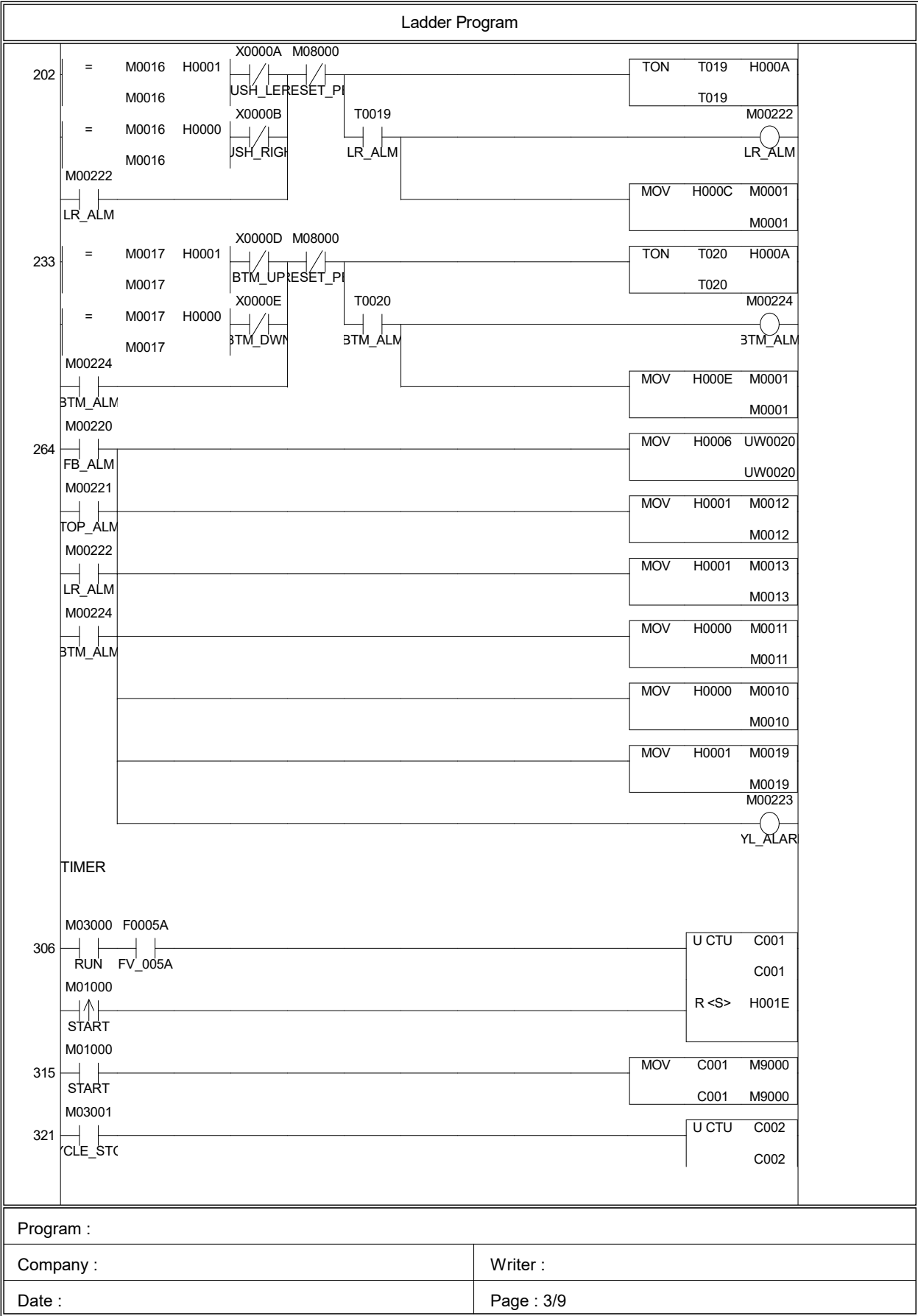
PARTS LIST		
ITEM No.	QTY	PART NAME
32	1	T PLATE 1
33	1	SUPPORT L1
34	1	SUPPORT L6
35	1	B COVER 3
36	1	HP PLATE
37	1	S PLATE
38	1	TX INPUT BLOCK 3
39	1	OP BLOCK TYPE 1
40	2	SIDE SB PLATE 4
41	1	R PLATE 1
42	1	T PLATE 3
43	1	L PLATE 2
50	1	COVER 3
51	1	ANTISTATIC COVER SIDE 2
52	1	ANTISTATIC COVER BOTTOM REAR
53	1	ANTISTATIC COVER BOTTOM
54	1	ANTISTATIC COVER BOTTOM FRONT
55	1	ANTISTATIC COVER 6.10.2020
56	1	SIDE SB PLATE 2
57	1	ANTISTATIC COVER SIDE 1
58	1	SBOX COVER 2
59	1	ANTISTATIC COVER 4 TOP REAR
60	1	LCD COVER A
61	1	LCD COVER 2
62	1	ANTISTATIC COVER BOTTOM SIDE
63	1	ANTISTATIC COVER REAR
64	1	PROTECT SCREEN
65	4	WHEEL BASE FOOT 4040

PARTS LIST			
ITEM No	QTY	PART NAME	DESCRIPTION
AE02	1	LP-S044-S1D-C5T-A	TOUCH PANEL DISPLAY
AE06	1	BTS30-LDTL	PHOTOELECTRIC SENSOR
AL01	4	AL 4040 1	AL FRAME 4040 L420
AL02	4	AL 4040 2	AL FRAME 4040 L840
AL03	4	AL 4040 3	AL FRAME 4040 L520
AL201	4	AL 2020 1	AL FRAME 2020 L300
AL204	2	AL 2020 4	AL FRAME 2020 L560
AL208	2	AL 2020 8	AL FRAME 2020 L297
AL209	1	AL 2020 9	AL FRAME 2020 L558
AL210	2	AL 2020 10	AL FRAME 2020 L315
AL211	2	AL 2020 11	AL FRAME 2020 L295
AL212	2	AL 2020 12	AL FRAME 2020 L300
AL213	2	AL 2020 13	AL FRAME 2020 L119
AP54	3	ACA1210-2	ABSORBER
DB44	1	L BRACKET 4040	L BRACKET 4040
DB55	2	CAP 2020 BLACK	END COVER 20x20
DB80	2	ANGLE 2020	ANGLE JOINT 2020
E01	2	GREEN SW	ON SWITCH
E04	1	KEY SWITCH	EMERGENCY SWITCH
E04	2	EMERGENCY SW S	EMERGENCY SWITCH
E06	1	FLIP ON SW	FLIP ON SWITCH
HM02	2	MGN9C1R 75	LINEAR SLIDE TABLE
OE45	2	OMRON GLS-S1 M2	SENSOR
SP03	2	JA15-5-080	FLOATING JOINT
SP116	1	CDJ2B16-45Z-M9B-B	PNEUMATIC ROD CYLINDER
SP19	1	KQ2L04-M5	L FITTING D4 M5
SP23	1	CDJ2B16-35Z-M9B-B	PNEUMATIC ROD CYLINDER
SP27	2	MXS20-20F-M9B	PNEUMATIC SLIDE TABLE
Z01	3	HINGE MERVIN	HINGE
ZE06	1	LDTM-G3	TOWER LIGHT
ZM02	1	HANDLE 96	HANDLE
ZM05	4	AC-300F	CASTER WITH LOCK

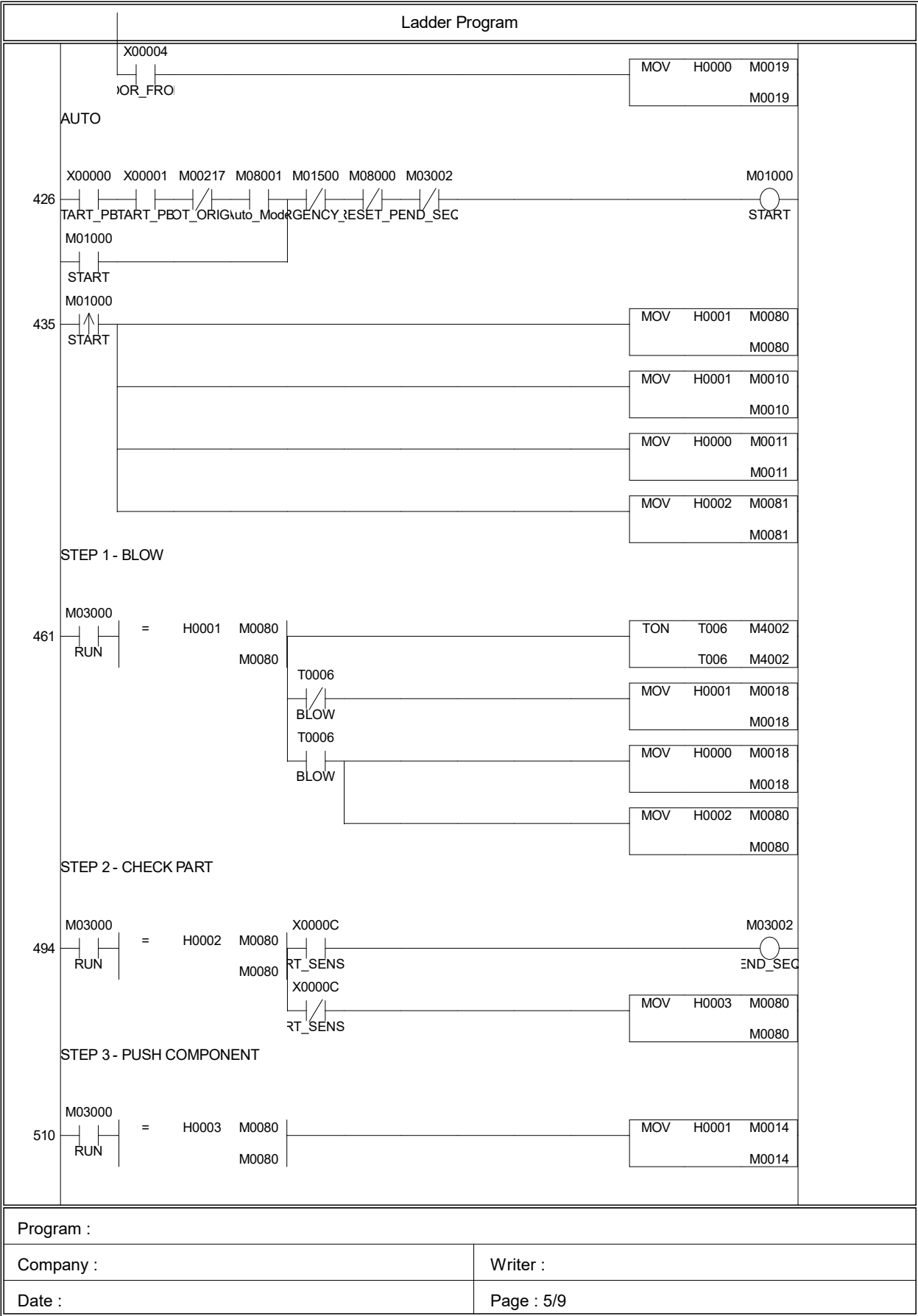
PLC I/O AND ELECTRICAL
WIRING DIAGRAM



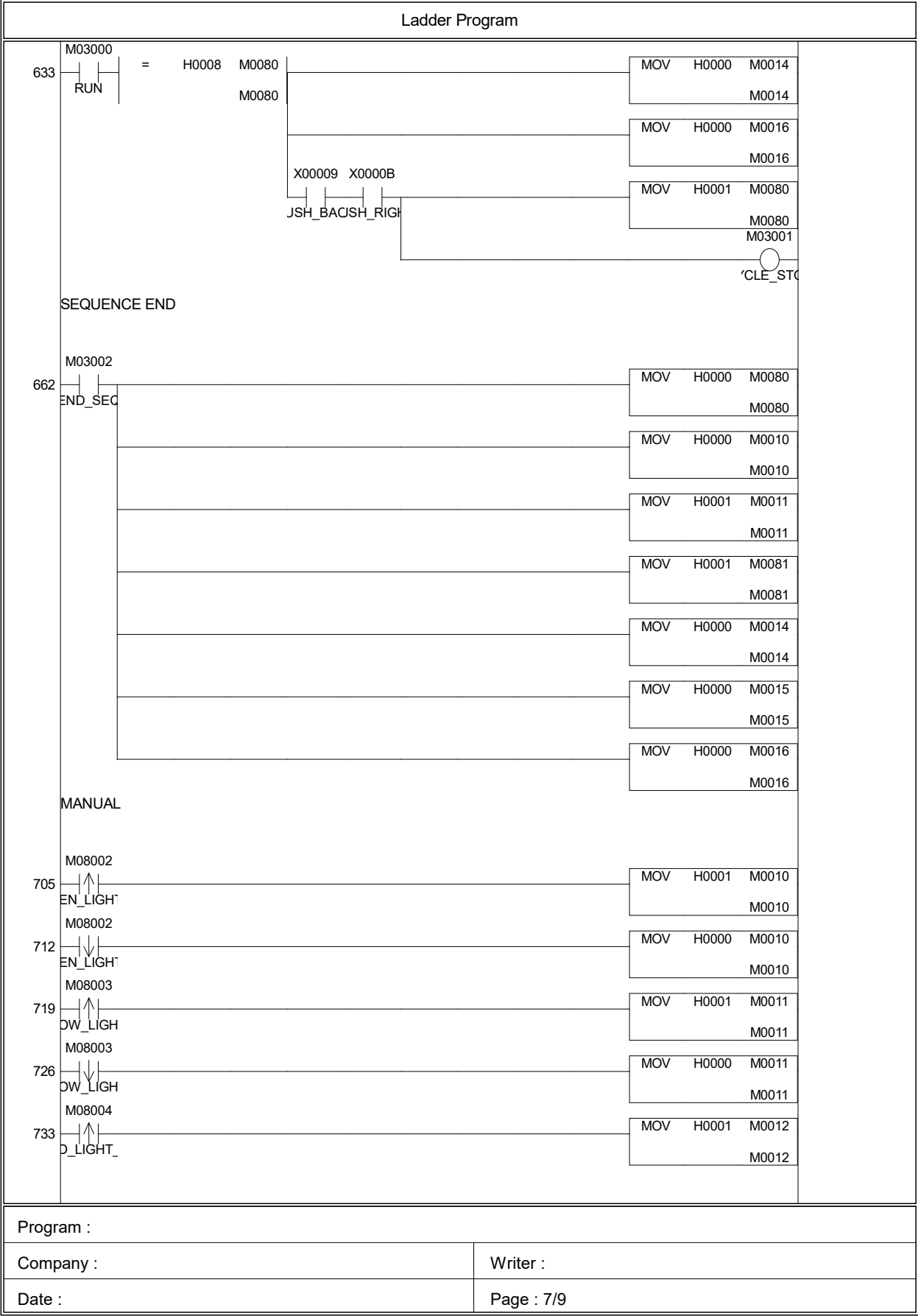




Ladder Program			
	M08011 ET_COUN F00010 328 FV_0010 X0000C M03001 334 RT_SENS'CLE_STC M01000 START X0000C RT_SENS F00010 345 FV_0010 RESET M08000 351 RESET_P	R <S> H270F MOV C002 M9004 C002 M9004 U CTU C003 C003 R <S> H270F MOV C003 M9005 C003 M9005 MOV H0000 M0010 M0010 MOV H0001 M0011 M0011 MOV H0001 M0081 M0081 MOV H0000 M0012 M0012 MOV H0000 M0013 M0013 MOV H0000 M0014 M0014 MOV H0000 M0015 M0015 MOV H0000 M0016 M0016 MOV H0000 M0017 M0017 MOV H0000 M0018 M0018 MOV H0002 UW0020 UW0020	
Program :			
Company :		Writer :	
Date :		Page : 4/9	



Ladder Program			
			MOV H0003 M0081 M0081
		X00008 ISH_FROI	MOV H0004 M0080 M0080
STEP 4 - HOLD COMPONENT			
535	M03000 RUN	= H0004 M0080 M0080	MOV H0001 M0015 M0015
		X00007 PUSH_DC	MOV H0005 M0080 M0080
STEP 5 - BEND PROCESS			
554	M03000 RUN	= H0005 M0080 M0080	MOV H0001 M0017 M0017
			TON T002 M4000 T002 M4000
		X0000D T0002 BTM_UP DELAY	MOV H0006 M0080 M0080
STEP 6 - BEND COMPLETE			
580	M03000 RUN	= H0006 M0080 M0080	MOV H0000 M0017 M0017
		X0000E BTM_DWN	MOV H0000 M0015 M0015
			MOV H0000 M0014 M0014
		X00006 X0000E TOP_UP BTM_DWN	MOV H0007 M0080 M0080
STEP 7 - PUSH PART			
614	M03000 RUN	= H0007 M0080 M0080	MOV H0001 M0016 M0016
		X0000A USH_LEF	MOV H0008 M0080 M0080
STEP 8			
Program :			
Company :		Writer :	
Date :		Page : 6/9	



Ladder Program			
740	M08004 D_LIGHT_	MOV H0000 M0012 M0012	
747	M08005 BUZZER	MOV H0001 M0013 M0013	
754	M08005 BUZZER	MOV H0000 M0013 M0013	
761	M08006 PUSH_PB	MOV H0001 M0014 M0014	
783	M08006 PUSH_PB	MOV H0000 M0014 M0014	
790	M08007 P_DW_PI	MOV H0001 M0015 M0015	
807	M08007 P_DW_PI	MOV H0000 M0015 M0015	
814	M08008 T_RIGHT	MOV H0001 M0016 M0016	
836	M08008 T_RIGHT	MOV H0000 M0016 M0016	
843	M08009 OTTOM_F	MOV H0001 M0017 M0017	
860	M08012 OTTOM_F	MOV H0000 M0017 M0017	
867	M08012 OWER_F	MOV H0001 M0018 M0018	
874	M08012 OWER_F	MOV H0000 M0018 M0018	
OUTPUT			
881	= M0010 H0001 M0010	Y00002 EEN_LIG	
887	= M0011 H0001 M0011	Y00001 LOW_LIG	
893	= M0012 H0001 M0012	Y00000 ED_LIGH	
899	= M0013 H0001 M0013	Y00003 ZER_VAL	
Program :			
Company :		Writer :	
Date :		Page : 8/9	

Ladder Program			
905	M00000 	Y00005 	
	M0014 	Y00004 	
	M0015 	ALVE_TO 	
	M0016 	Y00006 	
	M0017 	Y00007 	
	M0018 	VE_BOTT 	
	M0019 	Y00008 	
	INTERLOCK SENSOR 	BLOWER 	
	X00007 	Y00009 	
948	PUSH_DC 	MG_VALV 	
	X00008 		
	SH_FRO 		
	X0000A 		
	USH_LEF 		
	X0000D 		
	BTM_UP 		
953	X00000 X00001 M00217 M03000 	M00215 	
	TART_PBTART_PBOT_ORIG RUN 	ENCY_IN 	
958	= UW0020 H0002 	M00219 	
	UW0020 	JTO_PAG 	
964	M00215 		
	ENCY_IN 		
Program : Company : Date :			
		Writer : Page : 9/9	

MTV:- AUTO DIODE FORMING JIG

5 APRIL 2019

PLC Wiring

Model	Ref	Label	Description	Remark	Device Pin
LP-S044	COM1	24Vdc	24V Power supply		24Vdc
-S1D0(1)	COM2	24Vdc	24V Power supply		24Vdc
-C5T	+	24Vdc	24V Power supply		24Vdc
	-	0V	GND		0Vdc

Input List

Model	Ref	Label	Description	Remark	Device Pin
(INPUT1)	1	X0	START 1		
	2	X1	START 1		
	3	X2	EMG 1		
	4	X3	EMG 2		
	5	X4	DOOR FRONT		
	6	X5	DOOR BACK		
	7	X6	PUSH UP	UP/DOWN CYLINDER	
	8	X7	PUSH DOWN		
	9	X8	PUSH FRONT	FRONT/BACK CYLINDER	
	10	X9	PUSH BACK		
	11	XA	PUSH LEFT	LEFT/FRONT CYLINDER	
	12	XB	PUSH RIGHT		
	13	XC	PART SENSOR		
	14	XD			
	15	XE			
	16	XF			
	COM	24Vdc			
	COM	24Vdc			

Output List

Model	Ref	Label	Description	Remark	Location
(OUTPUT1)	1	Y0	RED	TOWER	
	2	Y1	YELLOW		
	3	Y2	GREEN		
	4	Y3	BUZZER		
	5	Y4	UP/DOWN	VALVE	
	6	Y5	FRONT/BACK		
	7	Y6	LEFT/RIGHT		
	8	Y7			
	9	Y8	EMERGENCY VALVE CUT_OFF	LATEST ADD 24MAC2025	
	10	Y9			
	11	YA			
	12	YB			
	13	YC			
	14	YD			
	15	YE			
	16	YF			
	COM	24Vdc			
	COM	0V			